

Hotel Booking Demand Analysis

Introduction

- Analyzes hotel booking demand and customer behavior
- Focuses on booking trends, cancellations, and pricing
- Uses EDA and machine learning for insights
- Helps improve decision-making and revenue

Business Problem

- High booking cancellations → revenue loss
- Poor resource utilization
- Difficulty in demand forecasting
- Inefficient pricing strategies
- Pricing not aligned with customer behavior
- Need to analyze booking patterns and cancellations

Problem Statement

- Understand booking patterns and customer behavior
- Reduce booking cancellations
- Improve revenue and operational efficiency
- Identify key factors affecting cancellations and pricing
- Enable data-driven decision-making

Objectives

- Analyze booking trends and customer behavior
- Identify factors influencing cancellations
- Study pricing using ADR
- Analyze impact of lead time
- Evaluate revenue by room types
- Build ML models for prediction
- Provide actionable business insights

Dataset

Key Features in the Dataset

- **Customer ID** – Unique identifier for each customer
- **Customer Country** – Country of the customer
- **Booking ID** – Unique booking reference

- **Hotel** – Type of hotel (City Hotel / Resort Hotel)
- **City** – Location of the hotel
- **Booking Date** – Date when booking was made
- **Arrival Date** – Date of check-in
- **Lead Time** – Number of days between booking and arrival
- **Adults, Children, Total Guests** – Number of guests
- **No. of Weekend Nights** – Weekend stay duration
- **No. of Week Nights** – Weekday stay duration
- **Total Nights Stay** – Total duration of stay
- **Meal Type** – Type of meal selected
- **Booking Status** – Target variable (0 = Not Cancelled, 1 = Cancelled)
- **Average Daily Rate (ADR)** – Price per night
- **Market Segment** – Booking source/category
- **Reserved Room Type** – Room type initially booked
- **Assigned Room Type** – Room type actually given
- **Special Request** – Number of special requests made by the customer

Methodology

- Data Collection (Pandas)
- Data Cleaning & Preprocessing
- Feature Engineering
- Exploratory Data Analysis (EDA)
- Data Visualization (Matplotlib, Seaborn)
- Machine Learning Models
- Model Evaluation
- Dashboard Development (Power BI)

Data Cleaning

- Handled missing values
- Converted date columns to date-time
- Checked categorical consistency
- Removed irrelevant data
- Validated numerical values
- Prepared target variable (Booking Status)

Feature Engineering

- Combined weekday and weekend nights to represent total stay duration.
- Calculated as $ADR \times Total\ Nights\ Stay$ to analyze financial contribution.

- Counts past cancellations per customer to capture behavioral patterns.
- Extracted from date columns to study seasonal trends.
- Combined adults and children to represent total number of guests.

Exploratory Data Analysis (EDA)

Univariate Analysis

- High cancellation rate observed
- Bookings split between city & resort hotels
- Online bookings dominate
- ADR mostly moderate
- Short lead time bookings are common

Bivariate Analysis

- Higher ADR → affects cancellations
- Higher lead time → more cancellations
- More special requests → less cancellations
- Past cancellations → repeat behavior
- Cancellation varies by location

Multivariate Analysis

- Cancellation varies by city & hotel type
- Standard rooms most booked
- Premium rooms generate higher revenue

Key Observations

- High lead time → high cancellations
- Customer behavior strongly impacts cancellations
- Special requests → higher commitment
- Premium rooms → higher revenue
- Pricing & location affect outcomes

Overall Conclusion (EDA)

- Lead time strongly affects cancellations
- Past behavior predicts future cancellations
- Special requests reduce cancellation risk
- Standard rooms popular, premium rooms profitable
- Location influences booking patterns

Machine Learning Model

- Target: Booking Status (0/1)
- Data split into training & testing

Models Used

- Logistic Regression → baseline
- Decision Tree → interpretable
- Random Forest → best performance

Evaluation

- Accuracy used for evaluation
- Random Forest performed best

Outcome

- Predicts high-risk bookings
- Supports proactive decision-making

Dashboard

- Interactive dashboard created
- Visualizes booking trends and cancellations
- Includes pricing and revenue insights
- Shows customer behavior patterns

Power BI Explanation

- Booking status overview (cancelled vs non-cancelled)
- Monthly trends in bookings and pricing
- ADR analysis and its impact on bookings
- Cancellation analysis
- Revenue by room type
- Geographical insights across cities and countries

Insights

- Online bookings have higher cancellations
- Most bookings are short stays (2–4 nights)
- Higher lead time → higher cancellations
- Past cancellations → repeat behavior
- Special requests → lower cancellations
- Premium rooms → higher revenue
- Pricing varies across seasons
- Cancellation varies by city & hotel type

Recommendations

- Improve cancellation policies
- Use dynamic pricing strategies

- Focus on short-stay customers
- Encourage special requests
- Optimize room allocation
- Target low-cancellation regions
- Use ML models for prediction

Final Conclusion

- Successfully analyzed booking behavior and trends
- Identified key factors affecting cancellations
- ML models enable prediction of cancellations
- Dashboard improves data visualization
- Helps reduce cancellations and optimize pricing
- Improves operational efficiency and revenue